Library Information Management System

Project Statement

Project Manager: Fardin Sahriar Al Rafat

Date: 2026-01-11

Approval Page

Library Inc, Client

Library Information Management System

LIMS

This document is to receive the approval of the client, Library Inc, and the project manager, Fardin Sahriar Al Rafat, for the proposed project. By signing this document, both parties acknowledge they have read the document in its entirety and agree to the plans in this document. If the proposed plan is accepted, date and sign below to move forward with the proposed project outlined in this document.

Approver Title	Approver Name	Signature	Date
Client	Peter MacDonald (Library Inc Dir)		

Project Manager	Fardin Sahriar Al Rafat		

Document Tracking

Version	Document Name	Date	Author(s)	Comments
1.0	CIS2261_ProjectStatement_Farhan_Fardin	2026.01.23	P.F S .Al'Rafat P. MacDonald	First Draft

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Project Summary

Summary:

Client	Library Inc
Project Name	Library Information Management System
Project Acronym	LIMS
Project Manager	Fardin Sahriar Al Rafat
Project Start Date	2026-01-05
Project Completion Date	2026-03-20
Total Budget Allocation	$23,500.00

Definitions:

Client	A corporate customer or end user that uses the system with limited privileges to search the library database. Additionally, with authorised privileges be able to view books loaned and make updates to the database.
Staff	A library staff member that uses the system without full administrative privileges but can create, read, update, and view catalog, client, and loan information according to their role.
Manager/Administrator	A staff member with full system access who can manage users, configure security roles, and maintain all library data and settings.
Database	An organized collection of the system data.
Company	Library Inc, the organization that owns and operates the LIMS solution.

Resources:

Microsoft Office Software
Draw.io
System Analysis and Database Documentation (ERD, Class Diagrams, Use Case, etc.).
XAMPP stack providing Apache, PHP, and MariaDB for local development and testing.
PHP IDE or editor (e.g., PhpStorm) for server-side development.
Business forms and processes documentation from the client
Human Resources (Project Manager, System Analysts, Developer, Client, etc.).

Project Objective

The objective of the Library Information Management System (LIMS) project is to replace Library Inc.'s manual or fragmented processes with a centralized web-based application that manages books, clients, suppliers, and loans more efficiently. The system will support day-to-day library operations and improve data accuracy, reporting, and access control for staff and management.

This project will implement LIMS as a PHP application using MySQLi with a MariaDB database running on XAMPP, following a multi-tier architecture that separates presentation (HTML/CSS/JavaScript), business logic (PHP), and data access (SQL). Users will access the system through a web browser, and role-based authentication implemented via PHP sessions will control access for Administrators and Staff, with future support for Client self-service.

The complete LIMS scope includes:

- **Catalog Management**
    - Add, update, delete, and view book records (title, type, publisher, etc.).
    - Manage supplier information.
    - Generate full book listings by type, title, and publisher, and custom listings by user-entered book type.
- **Loan and Overdue Management**
    - Record book check-outs and returns, including client, dates, and status.

- Track overdue items and generate overdue and weekly overdue reports.

- **Client Management**
  - Maintain client profiles (add, update, delete, view).
  - Produce lists of active clients as needed.
- **Security and Access Control**
  - Provide multi-level security for Administrators and Staff (and future Client access) to protect data and enforce role-based permissions.

Iteration One of this project will focus on delivering a working PHP/MariaDB prototype that supports core Catalog Management, Client Management, and basic Security features, forming the foundation for later Loan and Overdue modules. This undertaking is expected to be completed by March 9th, 2026, to allow time for staff training and active business usage by March 20th, 2026. The budget for this project is $23,500.00.

Project Deliverables

Phase	Deliverable	Description
1	**Elaboration & Preparation**	**Weeks 1-2**
	Business Meetings and Interviews	Meet (simulated) with Library Inc. management and library staff to refine requirements for catalog, client, and loan management, and confirm the use of a PHP/MariaDB web solution.
	Review Company Records	Examine existing book lists, client records, supplier details, and any loan tracking documents to understand current inputs, outputs, and workflows.
	Observe Business Operations	Analyze current processes for book check-out, returns, and overdue handling to identify pain points and automation opportunities.
	Review Project Statement	Finalize project scope, assumptions, constraints, key requirements, deliverables, timelines, budget, team roles, and standards to be followed.
2	**Analysis & Design**	**Weeks 3-6**
	System Overview	Provide a description of the overall LIMS, including objectives, deliverables, assumptions, constraints, key requirements, and high level system modeling (use case diagrams, event

		table, context diagram).
	System Component Details	A detailed design of the entire system, highlighting the output design, input design, database design, and any supporting components including forms or policies.
	Environmental Requirements	Hardware and software requirements, or any other necessary external modifications.
	Implementation Requirements	A description of any necessary data entry requirements, security considerations, access controls, training requirements, and postimplementation reviews.
3	**Development**	**Weeks 7-10**
	Database Scripts	SQL script to setup the database for the system.
	Source Code & Executable	Application and other code needed to run the system.
	Test Plan	Documentation to ensure all systems components meet requirements.
	Incident Report	Incidents or defects discovered during testing are logged and reported.
	Test Summary Report	Overview summary of the test results.
4	**Documentation**	**Week 10**
	Installation Guide	Instructions to install system.
	User Guide	Video demonstration to guide staff, stakeholders and clients through using the system.

Non-Deliverables

- Library Inc. will provide its own production hardware and long-term hosting infrastructure; deployment beyond the XAMPP based environment is out of scope for this iteration.

- Advanced loan features such as reservations, renewals, and automated email/SMS notifications will not be implemented in Iteration One.

- A full client-facing self-service portal (public search and online account management) is not part of Iteration One and may be considered in future iterations.

- Integration with external systems (e.g., third-party catalog services) and advanced analytics dashboards are not included in this project.

Assumptions

- Hardware and network infrastructure must meet minimum requirements to support the implemented system.
- Test equipment will be available as required.
- Users will be available to test and provide feedback for each stage of the development process.
- Project funding is sufficient to cover the entire project.
- The project scope or timeline will not change without written consent.
    - Appendix B: Change Request

Constraints

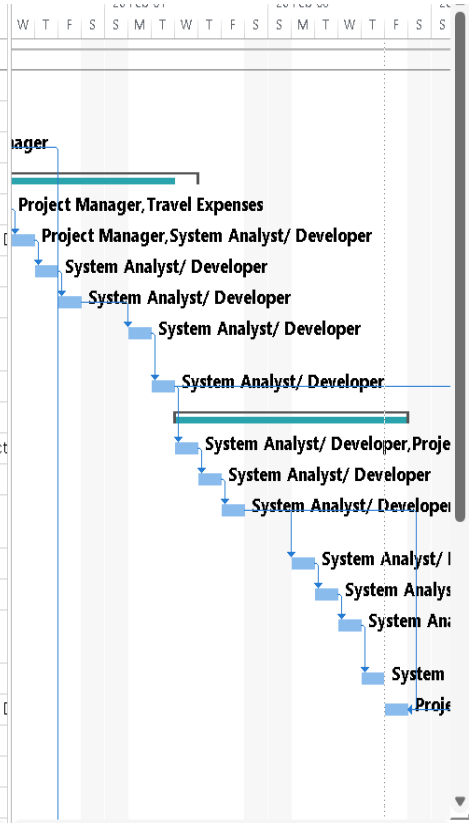
- Timeline: Project must be completed by March 20th, 2026, to allow for staff training and deployed for business usage by March 20th, 2026.
- Budget: Total project budget is fixed at 23,500.00. The Client and Project Manager must authorize any change to scope (budget and timeline).
- Technology Stack: The system must be implemented as a PHP web application using MySQLi and a MariaDB database running on XAMPP, in alignment with CIS technology guidelines.

Project Plan

Gantt Chart

GANTT CHART

	Task Mode	Task Name	Duration	Predecessors	Start	Finish	Resource Names
0		**Project LIMS Initiation**	**55 days**		**Mon 26-01-05**	**Fri 26-03-20**	
1		**Draft**	**15 days**		**Mon 26-01-05**	**Fri 26-01-23**	
2		Create Work Schedule/Gantt Chart	9 days		Mon 26-01-05	Thu 26-01-15	Project Manager
3		Create Project Statement	6 days	2	Fri 26-01-16	Fri 26-01-23	Project Manager
4		**System Analysis**	**8 days**		**Mon 26-01-26**	**Wed 26-02-04**	
5		Meet with Key Stakeholders	2 days	3SS	Mon 26-01-26	Tue 26-01-27	Project Manager,Travel Expenses
6		Event based analysis	1 day	5	Wed 26-01-28	Wed 26-01-28	Project Manager,System Analyst/ [
7		Creat domain class diagram	1 day	6	Thu 26-01-29	Thu 26-01-29	System Analyst/ Developer
8		Create single use case diagrams	1 day	7	Fri 26-01-30	Fri 26-01-30	System Analyst/ Developer
9		Create fully developed use case descriptions	1 day	8	Mon 26-02-02	Mon 26-02-02	System Analyst/ Developer
10		Create SSDs	1 day	9	Tue 26-02-03	Tue 26-02-03	System Analyst/ Developer
11		**System Design**	**8 days**		**Wed 26-02-04**	**Fri 26-02-13**	
12		Architectural Design	1 day	10	Wed 26-02-04	Wed 26-02-04	System Analyst/ Developer,Project
13		Create design class diagram	1 day	12	Thu 26-02-05	Thu 26-02-05	System Analyst/ Developer
14		Create Three-layered sequence diagram	1 day	13	Fri 26-02-06	Fri 26-02-06	System Analyst/ Developer
15		Create package diagram	1 day	14	Mon 26-02-09	Mon 26-02-09	System Analyst/ Developer
16		Create ERD	1 day	15	Tue 26-02-10	Tue 26-02-10	System Analyst/ Developer
17		Create Reports for selected output use cases	1 day	16	Wed 26-02-11	Wed 26-02-11	System Analyst/ Developer
18		Create UI for selected input use cases	1 day	17	Thu 26-02-12	Thu 26-02-12	System Analyst/ Developer
19		System Controls and Security	1 day	14FF	Fri 26-02-13	Fri 26-02-13	Project Manager,System Analyst/ [
20		**System Implementation**	**16 days**		**Mon 26-02-16**	**Mon 26-03-09**	
21		Database Implementation	2 days		Mon 26-02-16	Tue 26-02-17	System Analyst/ Developer
22		Coding	8 days	21	Wed 26-02-18	Fri 26-02-27	System Analyst/ Developer

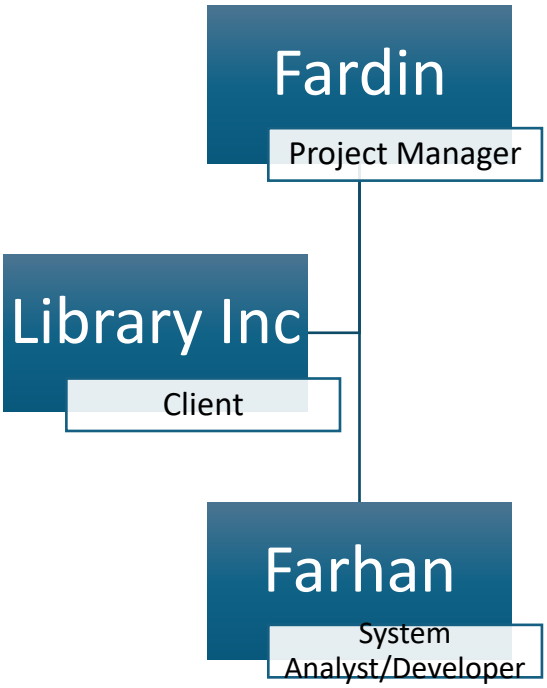


Library Information Management System

Resource Sheet

		Resource Name	Type	Material	Initials	Group	Max.	Std. Rate	Ovt. Rate	Cost/Use	Accrue	Base	Code	Add New Column
1		Project Manager	Work		PM		100%	$44.00/hr	$57.00/hr	$0.00	Prorated	Standard		
2		System Analyst/ Developer	Work		SA		100%	$40.00/hr	$57.00/hr	$0.00	Prorated	Standard		
3		Travel Expenses	Cost		TE						Prorated			

Team Organization



Responsibilities

Project Manager:

- Leads and plans the project.
- Meets with clients and stakeholders.
- Participate and supervise execution of each stage of the project.
- Time management and budget management.
- Evaluate client feedback and change requests.
- Documents about business rules and constraints.
- Help define system needs for book loans, overdue etc.

System Analyst/Developer:

- Conducts detailed system analysis and requirement modeling.
- Develops diagrams (Use Cases, Class Diagrams, DFDs, Event Table).
- Specifies system interactions and interface requirements.
- Ensures accuracy of technical modeling.
- Write and implement efficient code.
- Modifying system software to fix coding errors, adapt to new requirements, improve performance and/or security, or upgrade interfaces.
- Test and evaluate new system components.

Client:

- Communicate with the project manager to provide project details, goals, and authorization.
- Authorize change requests.
- Provide funding for the project.

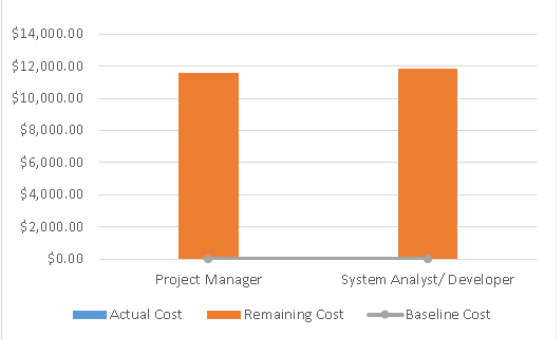
Budget

RESOURCE COST OVERVIEW

RESOURCE COST OVERVIEW

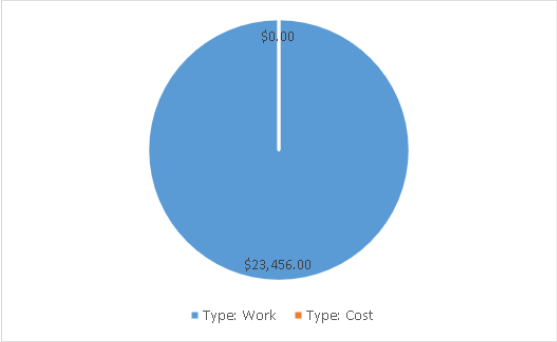
COST STATUS

Cost status for work resources.



COST DISTRIBUTION

How costs are spread out amongst different resource types.



COST DETAILS

Cost details for all work resources.

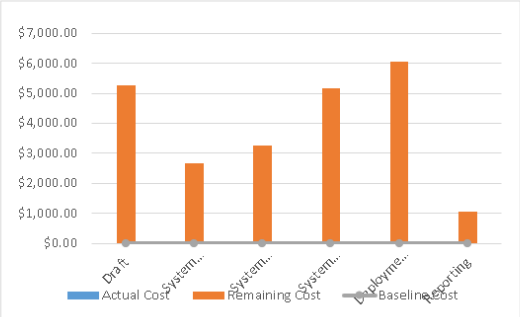
Name	Actual Work	Actual Cost	Standard Rate
Project Manager	0 hrs	$0.00	$44.00/hr
System Analyst/ Developer	0 hrs	$0.00	$40.00/hr

TASK COST OVERVIEW

TASK COST OVERVIEW

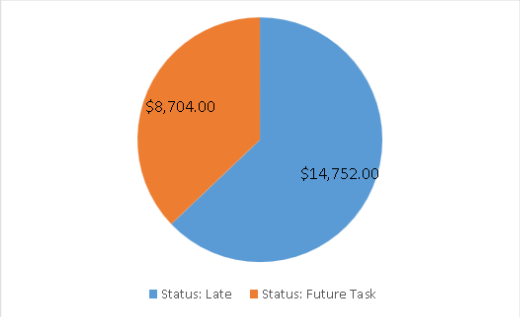
COST STATUS

Cost status for top-level tasks.



COST DISTRIBUTION

How costs are spread out amongst tasks based on their status.



COST DETAILS

Cost details for all top-level tasks.

Name	Fixed Cost	Actual Cost	Remaining Cost	Cost	Baseline Cost	Cost Variance
Draft	$0.00	$0.00	$5,280.00	$5,280.00	$0.00	$5,280.00
System Analysis	$0.00	$0.00	$2,656.00	$2,656.00	$0.00	$2,656.00
System Design	$0.00	$0.00	$3,264.00	$3,264.00	$0.00	$3,264.00
System Implementation	$0.00	$0.00	$5,152.00	$5,152.00	$0.00	$5,152.00
Deployment	$0.00	$0.00	$6,048.00	$6,048.00	$0.00	$6,048.00
Reporting	$0.00	$0.00	$1,056.00	$1,056.00	$0.00	$1,056.00

Change Management

Change management is the collection of processes, tools, and techniques that are used to manage project revisions to achieve new or altered requirements. Project management and change management support each other when moving a project from its current state, through a transition period, and into a future iteration. While project management will focus on the technical tasks necessary to achieve requirements, change management will focus on preparing and supporting project stakeholders through the change.

As a result, the project manager will be responsible for assessing feedback from all involved parties to determine the scope of any project changes. For small changes that will not affect the project scope (budget or timeline), the project manager will update the requirements accordingly and progress will continue immediately.

If a change will affect the scope of the project or alter important deliverables, the project manager will meet with stakeholders to identify the new requirements and the impact on project budget and scope. If it is agreed the change is necessary, a formal Change Request Form (Appendix B) must be completed and authorized by both the project manager and the stakeholders. The Change Request Form will contain the following:

- Key dates for the anticipated decision and expected implementation of the changes.
- A brief description of the changes.
- A description of the impact to the project timeline and budget
- Recommendations for a resolution.
- Approval or disapproval with signatures from the approvers.

Note: See Appendix B – Change Notification/Request Form for a template.

Progress/Status Reporting

Project status reports will be released on the schedule listed below. The project manager, Fardin Sahriar Al Rafat, will prepare the status reports and send them to the project stakeholders and the client, Library Inc.

Status reports will include the following:

- An overview of the progress made on the project.
- A list of outstanding issues and recommended actions.
- A list of the change requests.
- Plans for the next work period.
- An overview of the budget with an attached Microsoft Project file.

Schedule:

- Status Report One: January 30th, 2026
- Status Report Two: February 20th, 2026
- Status Report Three: March 6th, 2026

Note: See "Appendix C – Status Report Form" for a template of this form.

Decision Making

The decision-making process used throughout this project will consist of the following steps:

1. The problem or issue will first be identified.
2. Further information and details will then be gathered regarding this problem or issue, if necessary.
3. Solutions will be evaluated amongst team members and discussed.
4. Once the solutions have been discussed, the team will decide together which solution to implement to correct the problem or issue.
5. If there is not 100% agreement on a solution, the team will then discuss further until a decision can be made in which all members agree it is acceptable.
6. If the selected solution requires a significant change to the project scope, the project manager will meet with stakeholders to discuss the solution and identify any new requirements or any further impact to the project. If the stakeholders agree to the proposed solution a formal Change Request Form (Appendix B) will be completed and authorized by all parties involved.
7. The selected solution will be implemented and evaluated for effectiveness.

Note: To keep downtime to a minimum, all decisions will be made within 24 hours.

Standards & Conventions

Storage Units

A local desktop development environment using XAMPP will be used to develop the system components. Testing will be done in an environment with server infrastructure that is similar to the company production environment. The company's production infrastructure will be used to deploy the project systems.

Bitbucket and SourceTree utilizing Git will be used for version control of the source code, and for issue tracking.

Executable & Source Code File Naming

All executable and source code files will be names using the lowerCamelCase convention.

Source Code Internal Module Header Documentation

Each program file will include header documentation detailing the following:

Author(s):
Date:
Purpose:
Note:

Internal documentation will include single-line comments for small pieces of code requiring an explanation, while more detailed multi-line comments will be used for larger logical code blocks.

Database & Table Naming

The database name will use UpperCamelCase (e.g. Lims).

Table names will be in singular form, limited in length, and represent meaningful business entities written in UpperCamelCase (e.g. `Report`).

Field names will be singular and include the table name followed by an underscore and the attribute name in UpperCamelCase (e.g. `Report_Id`).

Variable / Function / Class Naming

Classes will be nouns written in UpperCamelCase (e.g. `ClassName;`).

Functions or methods will begin with verbs in lowerCamelCase (e.g. `getTotal();`).

All variables will be written in lowerCamelCase (e.g. `double total;`).

Constants will be written in uppercase characters with words separated by underscores (e.g. `MAX_TOTAL = 100;`).

Document Naming

Naming will follow the Holland College CIS Programming Standards format including the course number, project deliverable name, and all author last names. If multiple files are being submitted, all files will be included in a zip file.

CourseNumber_DeliverableName_AuthorLastNames

(e.g. CIS2261_ProjectStatement_Farhan_Fardin)

Appendices

Appendix A: Team Contract/Charter

Member Contact Information:

NAME	**EMAIL**	**PHONE**
Farhan	farhan@hollandcollege.com	902.439.0312
Fardin Sahriar Al Rafat	frafat@hollandcollege.com	902.314.0123

Communication:

- Microsoft Teams ○ Group text messaging ○ Video conferencing calls ○ Audio conferencing calls ○ Screen sharing ○ File sharing

Goals:

- To gain valuable experience that can be applied to future work.
  - Meet deliverable deadlines. ○ Stay within the project budget. ○ Establish milestones.
  - Learn from mistakes.
- Successfully create and deliver an analysis and design of the requested system that is:
  - Functional ○ Usable ○ Efficient ○ Simple
- What marks does the team as a whole want to achieve?
  - As a whole, the team would like to achieve a grade of 90% or higher.

Roles & Responsibilities:

1. Fardin Sahriar Al Rafat – Project Manager
2. Farhan – Analyst/Developer

Team Meetings:

- Group text messaging will occur daily to discuss and set plans for the next 24 hours.
- Audio or video conference call meetings will occur twice each week to ensure team members are on the same page and to check that progress is being made to meet deliverable deadlines.
- Meetings will be held using Microsoft Teams collaboration tools.
- Meeting agendas will be developed by first identifying any goals for upcoming deliverables. Team members will also provide input prior to the meeting to ensure all needs are fulfilled. Time spent on each topic will be estimated in advance and a plan will be developed accordingly.
- The project manager will start and facilitate meetings, but each topic covered during a meeting will be led by a designated team member who is focusing on the subject.
- Microsoft Teams will be used to schedule meetings, and the built in Microsoft Forms tool can be used to track decisions and activities through surveys, quizzes, and polls.

- Missing or being late for a meeting can lead to productivity loss, declined team morale, client dissatisfaction, and project management problems. To help avoid these issues meetings will be planned well in advance and all team members will be notified. Reminders and notifications will be set in Microsoft Teams.
- Some valid reasons for missing or being late to a meeting include poor planning or organization of a meeting. Scheduling conflicts are also a possibility. Since most communication and meetings will occur remotely through Microsoft Teams, technology failure is a risk.
- Repeated absences/lateness will be addressed by the project manager privately. An action plan will be developed to help the situation. This may include introducing time management lessons or utilizing a more flexible schedule.

Decision-Making:

Decisions on the development of this project will be made in consensus with the group conscience and will proceed as the group deems to be acceptable. However, decisions will be made in the following steps:

8. The problem or issue will first be identified.
9. Further information and details will then be gathered regarding this problem or issue, if necessary.
10. Solutions will be evaluated amongst team members and discussed.
11. Once the solutions have been discussed, the team will decide together which solution to implement to correct the problem or issue.
12. If there is not 100% agreement on a solution, the team will then discuss further until a decision can be made in which all members agree it is acceptable.

Note: Decisions will be made in 24 hours or less.

Decisions requiring assistance from the instructor will be those that may require further clarification or details in order to make a sound and efficient decision for the project.

Team Project Contribution & Member Accountability:

1. How will the team proceed in the event that someone's work is deemed unacceptable?

1.1. The issue will be addressed early and privately. This will help promote a more informal chat instead of a more serious discussion. It will also help prevent future occurrences of a similar issue.
1.2. Thoughtful questions will be asked to determine if the is a solvable problem that is causing the declined quality of work.
1.3. Relevant examples of more acceptable work will be provided.

2. If someone does not submit their work on time, what will be the course of action to ensure the project is completed and/or the overall team mark is not compromised?
    2.1. Questions will be asked to determine what happened, how the team member intends to recover, and how the team member intends to prevent it from happening again.
    2.2. Other team members will help complete the late task as quickly as possible to keep the project moving along and avoid compromising the team mark.
3. How will you reward individual members for outstanding contributions?
    3.1. Individual praise and showcasing their work.
    3.2. Overachieving members may receive time off or a more flexible schedule without compromising future deadlines.
4. Under what circumstances will you remove a team-member from your group?
    4.1. An excessive number of absences or missed deadlines.
    4.2. A team member leaves the CIS program. Access to project files and group collaboration tools will be revoked.
5. Grading assignment:
    5.1. The team would prefer to be graded as a group and receive the same marks.

Conflict:

Steps towards handling conflicts within the team are as follows:

1. Acknowledge the conflict.
2. Discuss the conflict and any possible resolutions that may be made.
3. If necessary, request feedback from another team member to assist in the resolution.

Revisiting & Revising the Team Charter:

The Team Charter will be re-evaluated using the decision-making protocols described in this Appendix. A review will occur at the mid-point of the project schedule, coinciding with Status Report Two on February 20th, 2026.

All project members and the client will receive signed copies of the original and revised Team Charter.

Appendix B: Change Notification/Request Form

LIMS | Change Request/Notification

DATE: REQUEST #:
TO:
FROM:
SUBJECT:

Key Dates			
Anticipated Decision:		Expected Implementation:	

Brief Description

Impact (Time)	Days:

Impact (Budget)	Amount:

Recommendation/Resolution

Approved ☐	Disapproved ☐ Reason:

Approver Title	Approver Name	Signature	Date
Project Manager			
Client			

Appendix C: Status Report Form

LIMS | Progress Status Report

DATE: REPORT #: TO:
FROM:
SUBJECT:

Overview of Progress

•

Outstanding Issues/ Recommended Action	Count: 0
1	

Change Requests Log	Count: 0
1	

Plans for Next Period
1.

Overview of Budget (attached MS Project file)

MS Project Link